

## Supplementary File S1 — STROBE / TRIPOD+AI-Style Reporting Checklist

This checklist adapts the **STROBE** statement (observational studies) and the **TRIPOD+AI** checklist (prediction-model reporting) to the cross-dataset machine-learning benchmarking design of this study. Items are mapped to the sections of ``paper_en_submission_v6.docx``.

| #  | Reporting item                                                            | Addressed in         | Status |
|----|---------------------------------------------------------------------------|----------------------|--------|
| 1  | Study design explicitly described (LODO/LOSO benchmarking)                | §1, §3.2             | Yes    |
| 2  | Data sources & eligibility, public repositories named                     | §3.1, §6.2           | Yes    |
| 3  | <b>Proxy (non-validated) labels</b> explicitly declared & derived         | §3.1                 | Yes    |
| 4  | Participant-level (not trial-level) partitioning to avoid leakage         | §3.2, §4.2           | Yes    |
| 5  | Handling of missing / degenerate data disclosed                           | §3.5, §4.4           | Yes    |
| 6  | No selective omission of underperforming datasets                         | §3.5                 | Yes    |
| 7  | Predictors / features described (EEG bands, windows)                      | §2, §3               | Yes    |
| 8  | Model specification (RF, EEGNet, calibration)                             | §3.3–§3.4            | Yes    |
| 9  | Model performance metrics (Acc, Macro-F1, BAcc, $\kappa$ ) with rationale | §3.2, Table 2        | Yes    |
| 10 | Uncertainty: 20-seed 95% CIs reported                                     | §4.1                 | Yes    |
| 11 | <b>Multiple-comparison control</b> : Holm–Bonferroni over 8 targets       | §4.1                 | Yes    |
| 12 | Significance testing (Wilcoxon signed-rank) described & corrected         | §4.1                 | Yes    |
| 13 | Model limitations (label collapse, class imbalance) discussed             | §4.4–§4.5, §5        | Yes    |
| 14 | Reproducibility: single-script regeneration, fixed seeds                  | §3.5, §6.1           | Yes    |
| 15 | Code availability (public repo + DOI plan)                                | §6.1                 | Yes    |
| 16 | Data availability (raw accessions + derived intermediates)                | §6.2                 | Yes    |
| 17 | Ethics / IRB exemption stated                                             | §3.7                 | Yes    |
| 18 | Funding / Competing interests / Author contributions declared             | Author Contributions | Yes    |

### Notes

- STROBE and TRIPOD+AI are reporting frameworks, not study-design mandates;
- All quantitative claims are reproducible from ``run_exp101_reproducible.py``